Mathematical Derivation and Experimental Validation of Excitation-Current Coordinate Reduction in Electrically Excited Synchronous Machines

Winding Factor Research Study

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Abstract

A common two-coordinate representation of the stator flux linkages of an electrically excited synchronous machine is investigated. The conventional shift $s = i_d + ki_e$ is exact only when both flux components are constant along its discarded current direction. We derive the differential conditions and distinguish them from constant-flux contour statistics. Three acquisition datasets are rebuilt with one stationary voltage-based processing procedure. Seven reduced models, from one constant factor to a bounded five-parameter current-dependent shift, are compared with a full three-current reference by shared-table reconstruction, independent shifted-surface comparison and nested whole-excitation-group prediction. The conventional model's joint prediction RMSE is 6.00–6.97 mVs. Two constant axis-shift factors reduce it to 5.49–6.53 mVs with two parameters, whereas the full three-current reference achieves 1.48–1.79 mVs on identical supported points. More flexible shifts have machine-dependent advantages and do not close this accuracy gap. Two constant axis factors provide a simple reduced-model compromise; retaining excitation is preferable when accuracy dominates memory. Derivative residuals, flux errors and numerical effects are interpreted separately. No complete permanent-magnet-machine equivalence or calibrated physical error floor is inferred.

1 Problem, evidence and scope

An electrically excited synchronous machine has three independently adjustable currents, whereas many compact stator-flux lookup tables have two current axes. The practical question is whether excitation can be absorbed into a shift of the stator-current coordinates without losing information about *both* stator flux linkages:

$$\psi_d = \psi_d(i_d, i_q, i_e), \quad \psi_q = \psi_q(i_d, i_q, i_e), \qquad s = i_d + ki_e. \tag{1}$$

Throughout this paper, "lossless" means only lossless with respect to these two functions. It does not mean a lossless physical machine. No additional magnetic scalar model is introduced.

The reduction conditions are derived explicitly below and evaluated on the available measurements.

We first explain the existing contour estimator, then derive exactness conditions. Their estimated violations motivate a hierarchy of restricted coordinate models. Two experiments address different questions: a shared-table reconstruction asks how much error remains after merging and storage; a whole-group prediction asks whether an unseen excitation level can be represented without fitting to it. A separate comparison between shifted original surfaces checks compatibility before merging. The complete per-group visual record is supplied in a companion atlas.

2 Data and stationary flux reconstruction

2.1 Three datasets and their limits

Set 1 contains 7,476 operating points in 13 nominal excitation groups from 2 to 40 A at a requested 3,000 rpm. Set 2 contains 8,092 points in 14 groups from 0 to 40 A at 3,000 rpm. Their recorded constants are $p = 3$, $R_s = 0.013\ \Omega$ and $R_e = 3.2\ \Omega$. These sets are treated as related measurement campaigns, not independent machines with established independent uncertainty.

Set 3 contains 9,022 points in ten nominal excitation groups from 3 to 30 A at 2,250 rpm. Its constants are $p = 4$, $R_s = 0.0366185\ \Omega$ and $R_e = 2.6\ \Omega$. It is a separate machine configuration. All parameters are identified separately; agreement of a procedure across machines would not imply transfer of the same numerical factor. The absence of an earlier flux export is addressed by processing the acquisition channels directly.

The saved input records already contain electrical-period averages of dq voltages and currents. They are not voltage waveforms sampled anew for this investigation. Repeated acquisitions are averaged arithmetically by file, requested speed, nominal excitation and operating-point index. Index-zero records indicate no assigned operating point; 284 such acquisition records are excluded from Set 3. The same averaging is used for every set, without a new outlier-selection rule. The three Set 1 files retain their separate acquisition identities before their operating points are pooled.

Nominal excitation labels define groups only. All transformations, flux calculations and derivative estimates use measured i_d, i_q, i_e. Tracking differences, measured excitation ranges, repetition counts, speed ranges and temperature channels are exported per group. Set 1/2 stator temperature channels span approximately 50–80 °C; Set 3 spans approximately 40–104 °C. Set 3's recorded rotor-temperature channel includes negative values near −85 °C and is not accepted as a calibrated rotor-temperature history. No temperature correction is inferred from these channels. A current-only map can therefore also absorb variation due to unmodelled temperature or acquisition effects.

After repetition averaging, mechanical-speed telemetry spans approximately 2,934–3,072 rpm in Set 1, 2,933–3,071 rpm in Set 2 and 2,247–2,252 rpm in Set 3. The flux calculation uses the separately recorded electrical angular-speed channel. Requested speed does not replace this measurement. The recorded current-sensor delays also differ between Sets 1 and 2; this is an acquisition difference, not an identified cause of a reduction error.

2.2 Conventions and source of the flux values

The stationary dq convention is

$$u_d = R_s i_d + \dot{\psi}_d - \omega_e \psi_q, \qquad u_q = R_s i_q + \dot{\psi}_q + \omega_e \psi_d, \tag{2}$$

$$\omega_e = p\omega_m. \tag{3}$$

At nonzero steady speed, the source flux values are calculated as

$$\psi_d = \frac{u_q - R_s i_q}{\omega_e}, \qquad \psi_q = \frac{R_s i_d - u_d}{\omega_e}. \tag{4}$$

Measured electrical angular speed and measured dq voltages are used, rather than requested speed or controller voltage commands. The stored stator-resistance channel supplies R_s. The signs and source channels agree with the current corrected evaluation source. Its differential-inductance assignment was also inspected: derivatives of ψ_d with respect to i_q are L_{dq}, while derivatives of ψ_q with respect to i_d are L_{qd}. Older saved inductance arrays are not reused as measurements of these derivatives.

The dq channels follow the existing amplitude-based machine convention; they are not reinterpreted as phase RMS values. In Set 1/2, the median ratio $\sqrt{i_d^2 + i_q^2}/(\sqrt{2} I_{\mathrm{phase,rms}})$ is approximately 0.995,

consistent with this convention within acquisition differences. Set 3 lacks that RMS comparison channel and the recorded phase-selector metadata, so its amplitude convention is inherited from the same dq channel definitions rather than independently verified. A conventional three-phase amplitude-invariant torque formula is used only in Section 8 to explain coordinate substitution. This study does not independently calibrate the Park transformation, sensor gains, timing, resistance, inverter voltage or rotational losses. Agreement with (4) establishes arithmetic consistency, not absolute flux accuracy. Flux is expressed in Vs, with reported errors in mVs; 1 Vs = 1 Wb for these linkage units.

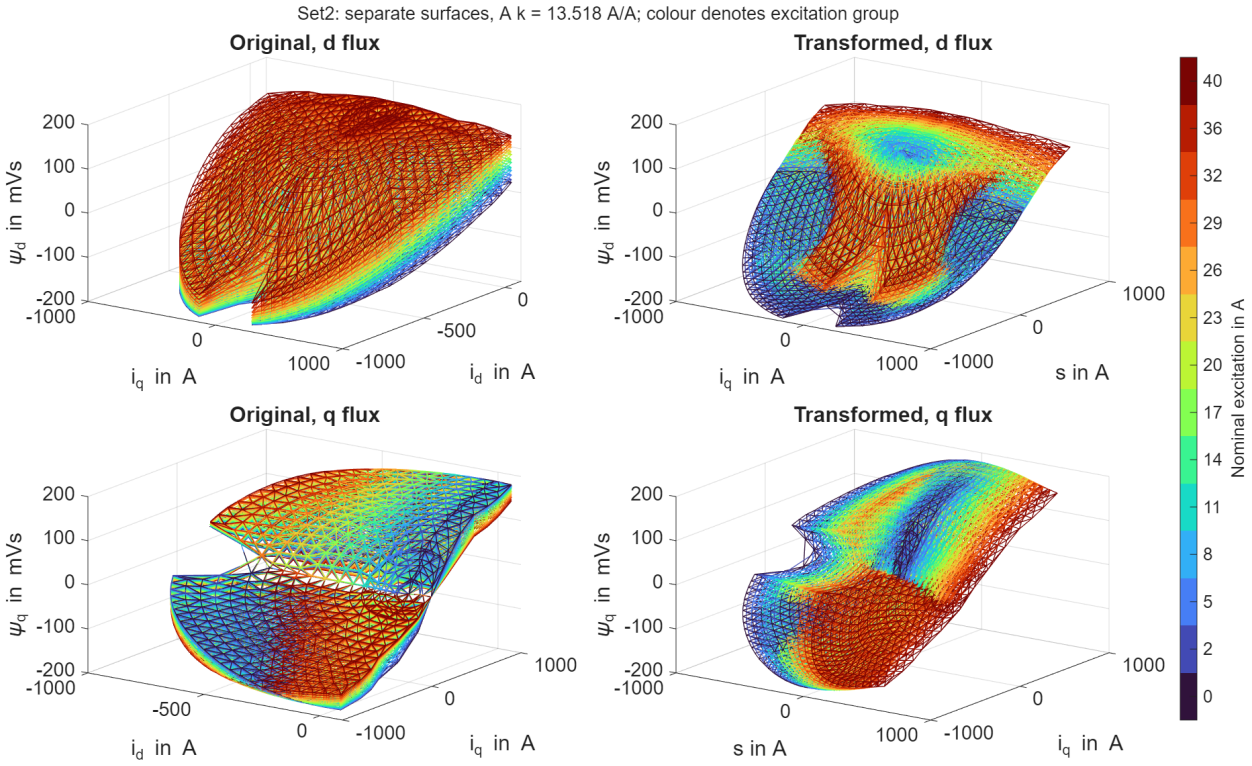


Figure 1: Original and independently transformed Set 2 meshes using the conventional factor. Colour identifies excitation groups. Open meshes keep overlapping excitation levels visible. The transformation changes coordinates, not source flux values. Separate transformed surfaces have not yet been replaced by one shared table.

3 What the conventional contour factor estimates

3.1 A local constant-flux displacement

Define all differential inductances with the output index first:

$$L_{xy} = \frac{\partial \psi_x}{\partial i_y}, \qquad x \in \{d, q\}, \quad y \in \{d, q, e\}. \tag{5}$$

A partial derivative changes one input and holds the other two fixed. Thus L_{de} measures the change in d-flux due to excitation with both stator currents fixed; it is not a derivative along an arbitrary contour-search path. Assuming continuously differentiable fluxes,

$$\mathrm{d}\psi_d = L_{dd}\mathrm{d}i_d + L_{dq}\mathrm{d}i_q + L_{de}\mathrm{d}i_e. \tag{6}$$

On a constant-ψ_d path, with $L_{dd} \neq 0$, the implicit-function theorem gives

$$-\frac{\mathrm{d}i_d}{\mathrm{d}i_e} = \frac{L_{de}}{L_{dd}} + \frac{L_{dq}}{L_{dd}}\frac{\mathrm{d}i_q}{\mathrm{d}i_e}. \tag{7}$$

At fixed i_q, only the first term remains. If i_q moves, cross-coupling contributes the second term. This is one possible bias in interpreting a contour statistic as a fixed-i_q derivative. It is not a demonstration that all observed errors come from i_q drift.

For finite endpoints on the same differentiable path,

$$-\frac{\Delta i_d}{\Delta i_e} = \frac{1}{\Delta i_e} \int_{i_{e,1}}^{i_{e,2}} \left(\frac{L_{de}}{L_{dd}} + \frac{L_{dq}}{L_{dd}} \frac{\mathrm{d}i_q}{\mathrm{d}i_e} \right) \mathrm{d}i_e. \tag{8}$$

A secant is a path average. It becomes a pointwise derivative only in a suitable limit. The assumptions fail where the implicit path is singular, disconnected or cannot be parameterized by i_e.

3.2 The actual averaging procedure

The conventional estimator orders complete groups by increasing nominal excitation. Within each group, it triangulates measured (i_d, i_q) points, rejects unusually long edges, and finds edge intersections with 100 d-flux levels spanning the middle 25–75% of the pooled d-flux value range. All three measured current coordinates (i_d, i_q, i_e) are interpolated at these intersections. Their arithmetic coordinate means are used; they are not arclength-weighted geometrical contour centroids. Adjacent-group ratios are then formed:

$$k_A = \mathrm{mean}_{g,\ell} \left[\frac{|\bar{i}_{d,g+1,\ell} - \bar{i}_{d,g,\ell}|}{\bar{i}_{e,g+1,\ell} - \bar{i}_{e,g,\ell}} \right]. \tag{9}$$

Only finite, numerically nonzero ratios enter the arithmetic mean. The absolute numerator removes directional information. Excitation ordering matters because the denominator retains its sign. The present implementation follows this rule rather than assuming an unsigned denominator or a geometrical turns ratio. The interactive plots additionally evaluate 201 levels over the full pooled d-flux range for display. These additional levels do not enter k_A: its original 100 levels and aggregation rule are unchanged. The shaded interval denotes fractions of the flux value range, rather than quantiles of the measurement population.

All intersection points lie on the interpolated contour, but their coordinate centroid generally does not: for a nonlinear function, evaluating it at the mean coordinate is not equivalent to averaging its values. Consequently, even (8) need not apply to the two selected centroids. Their mean i_q can also differ between groups. Triangulation, contour-level choice, sample density, coverage, edge filtering and the unweighted average of ratios all affect k_A. Points near densely intersected edges receive a different effective influence from points in sparsely sampled parts of a contour. This geometry can move both $\bar{i}_q$ and $\bar{i}_d$, so it can directly change the numerator of (9). Incomplete contours can contribute to this displacement. Whether these mechanisms explain a particular measured asymmetry requires a separate support and discretization investigation; symmetry is not imposed. Excluding the outer flux quarters during estimation does not validate applying the result there.

For a level λ, the plotted q-current drift is

$$D_g(\lambda) = \frac{\bar{i}_{q,g+1}(\lambda) - \bar{i}_{q,g}(\lambda)}{\bar{i}_{e,g+1}(\lambda) - \bar{i}_{e,g}(\lambda)}, \qquad \epsilon_g(\lambda) = \widetilde{\psi}_{d,g}(\bar{i}_{d,g}, \bar{i}_{q,g}) - \lambda, \tag{10}$$

where $\widetilde{\psi}_{d,g}$ is the original group's interpolated d-flux surface. The departure ϵ_g is a contour-construction diagnostic, not a shared-table reconstruction error. It can be nonzero even when an exact coordinate reduction exists.

The interactive 3-D scatter diagnostic places λ on its first axis and

$$Q_g(\lambda) = \tfrac{1}{2}\big(\bar{i}_{q,g}(\lambda) + \bar{i}_{q,g+1}(\lambda)\big) \tag{11}$$

on its second axis. Its third axis is the existing absolute-numerator contour ratio or D_g, and its colour is the mean of the two measured-excitation centroids. Thus Q_g is a mean of two already averaged q-coordinates; it is not a query at fixed physical i_q, need not span the measured q-current range and need not be symmetric. No local field $k(\psi_d, i_q)$ is identified by this display.

For a generic ratio $r = A/B$, first-order uncertainty propagation is

$$\delta r \simeq \frac{\delta A}{B} - \frac{A\,\delta B}{B^2}, \qquad \mathrm{var}(r) \simeq \frac{\sigma_A^2}{B^2} + \frac{A^2\sigma_B^2}{B^4} - \frac{2A\,\mathrm{cov}(A,B)}{B^3}. \tag{12}$$

The inverse powers of B explain why a small excitation difference or small inductance denominator makes a ratio sensitive. If the denominator vanishes, the desired factor may be unidentifiable rather than merely large. Unweighted averaging does not remove this sensitivity.

Method B below keeps the query i_q fixed while aligning d-flux cuts through a shared table. It avoids the moving contour centroid. It still has sampling and interpolation effects, and optimizing d-flux alone does not establish a common reduction of both components.

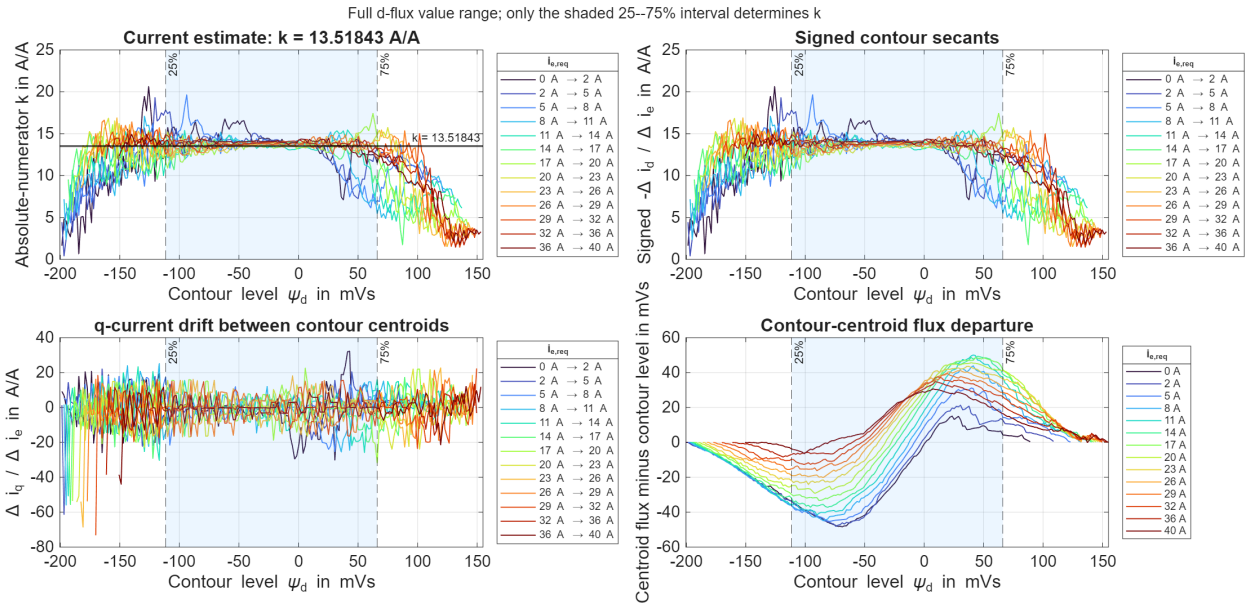


Figure 2: Set 2 contour statistics. Each curve represents a group or adjacent-group pair. The signed and absolute-numerator secants differ only when the d-displacement changes direction. The lower panels show q-current drift and the departure of contour centroids from the selected d-flux level. These centroids are not constant-flux samples at fixed q-current.

4 Exact reduction: local and global statements

Let $\Omega \subset \mathbb{R}^3$ be open, let $\boldsymbol{\psi} = (\psi_d, \psi_q)^\mathsf{T} \in C^1(\Omega)$, and define

$$\pi_k(i_d, i_q, i_e) = (i_d + k i_e, i_q) = (s, q). \tag{13}$$

The reduction is exact if a single-valued C^1 function $\boldsymbol{F}$ exists with

$$\boldsymbol{\psi}(\boldsymbol{i}) = \boldsymbol{F}(\pi_k(\boldsymbol{i})). \tag{14}$$

In plain terms, any two supported operating points with the same reduced coordinates must demand the same *pair* of flux values.

Proposition 1. *For a prescribed constant k, exact factorization implies*

$$L_{de} = kL_{dd}, \qquad L_{qe} = kL_{qd}. \tag{15}$$

Conversely, these identities imply local factorization. They imply global factorization on $\pi_k(\Omega)$ if every nonempty fibre of π_k in Ω is connected. More generally, disconnected fibre components must additionally have equal flux values.

Proof. At fixed (s, q) write $i_d = s - ki_e$. The chain rule gives

$$\left.\frac{\partial \boldsymbol{\psi}(s - ki_e, q, i_e)}{\partial i_e}\right|_{s,q} = \begin{bmatrix} L_{de} - kL_{dd} \\ L_{qe} - kL_{qd} \end{bmatrix}. \tag{16}$$

If (14) holds, varying the discarded coordinate cannot change either output, so both entries vanish. Conversely, zero derivative makes each output constant on each connected interval of a fibre. The map $(i_d, i_q, i_e) \mapsto (s, q, i_e)$ is an invertible linear coordinate change. Locally, the flux therefore does not depend on its third coordinate. Connected fibres make the local values agree globally. Because the projection is a submersion, these locally defined C^1 functions descend to the open reduced domain. □

Connectedness of the full domain alone is insufficient if a fibre has separated components. No nonzero L_{dd} is needed for the proposition itself; that assumption was needed for the contour slope. To identify a unique factor from the two flux derivatives, however, the sensitivity vector below must be nonzero. If both sensitivity and excitation-coupling vectors vanish, any k satisfies the local constraint. If only the sensitivity vanishes, a nonzero excitation coupling cannot be compensated by a d-shift.

4.1 Determinant, direction and joint regression

At one operating point define

$$\boldsymbol{a} = \begin{bmatrix} L_{dd} \\ L_{qd} \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} L_{de} \\ L_{qe} \end{bmatrix}, \quad \Delta = L_{dd}L_{qe} - L_{qd}L_{de}. \tag{17}$$

The columns describe the two-output response to d-current and excitation current. The determinant is their signed parallelogram area, in H^2. If $\boldsymbol{a} \neq 0$, $\Delta = 0$ means that $\boldsymbol{b}$ is parallel or antiparallel to $\boldsymbol{a}$: some local scalar factor can compensate excitation. It does not establish that this factor is constant throughout the domain. The normalized measure

$$\eta = \frac{|\Delta|}{\|\boldsymbol{a}\|\|\boldsymbol{b}\|} = |\sin\theta| \tag{18}$$

is dimensionless and undefined when either vector vanishes. It quantifies directional disagreement, not a percentage of flux error. A value of 0.1 is not a validated 10% flux error.

For a positive weight matrix W, the local weighted least-squares factor and the global constant counterpart are

$$k_{\mathrm{loc}} = \frac{\boldsymbol{a}^{\mathsf{T}} W \boldsymbol{b}}{\boldsymbol{a}^{\mathsf{T}} W \boldsymbol{a}}, \qquad k_{\mathrm{glob}} = \frac{\sum_j \boldsymbol{a}_j^{\mathsf{T}} W \boldsymbol{b}_j}{\sum_j \boldsymbol{a}_j^{\mathsf{T}} W \boldsymbol{a}_j}. \tag{19}$$

Here $W = I$ is used. The global factor weights locations with large d-current sensitivity more strongly; it is not generally the arithmetic mean or median of local factors. The derivative residual $\boldsymbol{b} - k\boldsymbol{a}$ has units H. Its relative aggregate norm

$$\rho(k) = \frac{\left(\sum_j \|\boldsymbol{b}_j - k\boldsymbol{a}_j\|^2\right)^{1/2}}{\left(\sum_j \|\boldsymbol{b}_j\|^2\right)^{1/2}} \tag{20}$$

is dimensionless. Neither quantity is a flux RMSE. Integrating a derivative residual along a specified finite path would require its location, direction and path length; a norm alone does not provide the resulting flux error. Minimizing ρ therefore need not minimize reconstruction or prediction RMSE.

4.2 How the derivatives are estimated

At each measured current point we fit both flux components to separate local affine functions of the three currents. Current coordinates are range-normalized. The nearest 40, 80 or 120 points enter a Gaussian-distance-weighted least-squares fit; the primary count is 80. A neighborhood must contain at least two excitation groups, and ill-conditioned local designs are omitted. The fitted slopes are converted back to H. Low d-sensitivity vectors are excluded from joint-factor identification, and small component denominators are excluded from the separate ratios. The thresholds are numerical conditioning rules relative to the median sensitivity, not physical validity bounds.

Local regression estimates a slope from a neighborhood and generally does not interpolate every sample. A radial-basis interpolant instead forms a weighted sum of distance kernels, with coefficients typically obtained from a global interpolation system; it is neither this local affine regression nor an independent measurement of inductance. No RBF replacement is used here. Neighbor count, noise, curvature and uneven sampling affect all reported derivatives. The sensitivity comparison uses common retained derivative points across neighborhood counts.

5 Restricted improvements and their mathematical limits

5.1 Constant and integrated shifts

Methods A, B and C all use $s = i_d + k i_e$, but estimate k differently. A is (9). B minimizes d-flux prediction error at the observed query i_q, with no contour-centroid motion. C minimizes prediction error of both flux components. C thus optimizes a flux criterion rather than (19). It is a numerical minimum within a stated bounded search, not a proof of a global optimum over all coordinate functions.

For $s = i_d + a(i_e)$, exactness requires $\boldsymbol{b} = a'(i_e)\boldsymbol{a}$. A differential factor is a local slope, while a is the accumulated shift:

$$a(i_e) = \int_0^{i_e} k_{\text{diff}}(\xi)\mathrm{d}\xi, \qquad a(0) = 0. \tag{21}$$

In general $a(i_e) \neq i_e k_{\text{diff}}(i_e)$. The zero-excitation reference is a coordinate gauge and does not claim that zero excitation was measured in every dataset.

Using fixed current scales I_d, I_q, I_e determined from the training currents, define $D = i_d/I_d$, $Q = i_q/I_q$, $E = i_e/I_e$. We fit a deliberately small hierarchy:

$$\text{D: } a(i_e) = i_e(c_1 + c_2 E), \tag{22}$$

$$\text{E: } a(i_q, i_e) = i_e(c_1 + c_2 E + c_3 Q^2), \tag{23}$$

$$\text{G: } a(i_d, i_q, i_e) = i_e(c_1 + c_2 E + c_3 Q^2 + c_4 D + c_5 D^2). \tag{24}$$

All c_j have units A/A. The chosen even q-dependence is a parsimonious saturation approximation, not an imposed exact machine symmetry. The number of coefficients is fixed before examining the final prediction groups. D is a continuous one-dimensional shift, not one independent translation per group. Its differential slope is $c_1 + 2c_2 E$. E adds a continuous variation within each excitation group. Its local excitation slope is $c_1 + 2c_2 E + c_3 Q^2$, whereas the total shift is (23).

For E, the identities $\boldsymbol{b} = a_e \boldsymbol{a}$ remain necessary. Its dependence on q changes the reduced q-gradient but does not remove the requirement for a common d-compensation direction for both flux components. If that direction does not exist, fitting more values of a pure d-shift cannot arbitrarily remove the incompatibility.

5.2 A shift of both stator-current axes

Method F introduces two constants in the *same* coordinate transformation for both outputs:

$$s_d = i_d + k_d i_e, \qquad s_q = i_q + k_q i_e. \tag{25}$$

The chain rule gives the necessary condition

$$\begin{bmatrix} L_{de} \\ L_{qe} \end{bmatrix} = \begin{bmatrix} L_{dd} & L_{dq} \\ L_{qd} & L_{qq} \end{bmatrix} \begin{bmatrix} k_d \\ k_q \end{bmatrix}. \tag{26}$$

The same fibre argument supplies local sufficiency, with corresponding global fibre conditions. An invertible stator Jacobian allows a local vector of compensating slopes, but it does not imply that the same constants work everywhere. F is distinct from assigning separate d-axis shifts to ψ_d and ψ_q. That latter construction uses different coordinates for different outputs and is not investigated here.

5.3 The bounded three-current extension

G retains two reduced axes, $s = i_d + a(\boldsymbol{i})$ and $q = i_q$, while allowing its shift to depend on all three currents. Define $k_{\mathrm{map}} = a/i_e$ by (24); at $i_e = 0$ its continuous formula is used. This is a transformation parameter, not automatically k_{diff}. The full chain rule is

$$\dot{s} = (1 + i_e k_{\mathrm{map},i_d})\dot{i}_d + i_e k_{\mathrm{map},i_q}\dot{i}_q + (k_{\mathrm{map}} + i_e k_{\mathrm{map},i_e})\dot{i}_e. \tag{27}$$

At fixed (q, i_e) its inverse exists locally when

$$s_{i_d} = 1 + \frac{i_e c_4}{I_d} + \frac{2 i_e c_5 i_d}{I_d^2} \neq 0. \tag{28}$$

The search enforces $s_{i_d} \geq 0.25$ on the rectangular training-current box. Since (28) is bilinear in i_d, i_e, checking its corner values bounds it throughout that box. Thus the d-coordinate is strictly monotone there for fixed other currents. Exact factorization still requires

$$\boldsymbol{b} = \frac{a_e}{1 + a_{i_d}} \boldsymbol{a}. \tag{29}$$

Nonparallel sensitivity vectors cannot be made parallel by this additional scalar freedom. The five-parameter form is a complexity comparison, not a construction capable of independently encoding every measurement. On an empirical group surface with varying measured $i_e(i_d, i_q)$, its projected triangles can nevertheless fold; this is a different check from (28) and is reported separately.

5.4 Full three-current reference and complexity

H uses (i_d, i_q, i_e) directly, with no coordinate-reduction assumption. It retains an excitation axis and therefore a larger table. It is an accuracy reference rather than exact ground truth: it still depends on measurements, interpolation and storage. A–C have one fitted shift parameter; D and F have two; E has three; G has five; H has no shift parameter. All reduced methods store two 81×81 flux arrays. H stores two arrays with 81 nodes on each stator-current axis and a bounded excitation-axis node count. Actual node counts and double-precision flux storage are recorded with the results; finite-support masks and axis vectors add a small implementation-dependent overhead.

6 Experiments and fair comparison

6.1 Source values, surfaces, shared interpolant and stored raster

Four objects must be kept distinct: (i) fluxes calculated from averaged measurement channels; (ii) separate piecewise-linear original surfaces, one per nominal group; (iii) a shared scattered field after transformation; and (iv) a stored regular-grid lookup table sampled from that shared field. Original group surfaces also interpolate measured excitation as a third scalar output. A group therefore represents an excitation *graph*, not an exactly constant measured-excitation plane.

For each model, every original measured current is transformed. A joint Delaunay triangulation is built in reduced coordinates, using all retained groups, and the two flux components are interpolated linearly on the same simplices. Exact coordinate duplicates are averaged. The field is sampled onto the common storage grid and queried again at each transformed physical operating point. In particular,

$$\widehat{\psi}_x(i_d, i_q, i_e) = \widehat{F}_x(i_d + ki_e, i_q), \qquad r_x = \widehat{\psi}_x - \psi_{x,\text{source}} \tag{30}$$

for A–C, with the corresponding coordinate map for the other methods. A shared approximation is actually constructed and queried; no identity obtained merely by shifting coordinates and shifting them back is counted as evidence.

An interpolant passing through distinct transformed training nodes can reproduce all training values even when nearby excitation surfaces disagree. Such interpolation identity does not establish physical exactness of the reduction. Raster reconstruction deliberately evaluates the shared representation after a finite storage operation; the independent-surface and missing-group experiments address additional questions.

Axes are normalized by their training ranges before triangulation. A simplex is rejected if any squared edge length exceeds nine times the mean squared length of the unique edges. Evaluation outside the retained support is missing, and a raster cell touching a missing node remains unavailable. This is a stated numerical support rule, not a calibrated physical domain. It avoids filling large unsupported gaps with artificial surfaces. Primary comparisons normalize both the 2D and 3D coordinates; a raw-coordinate control is also run for both dimensions.

Every reduced model uses 81 nodes per reduced axis. H uses 81 nodes per physical stator axis and between five and 21 excitation nodes, based on distinct rounded measured excitation values. Its realized dimensions are reported. A separate 41/81/161-node study changes only the rasterization of A. It is a numerical control, not a proposed remedy for a deficient coordinate model.

Saved historical single-group tables require a separate coordinate convention. If a physical slice has merely been relabelled by a fixed reference offset ke_g, its proper query coordinate is $i_d + ke_g$, even when the measurement at that point has a different excitation. Applying a pointwise measured-current offset to that relabelled slice mixes two coordinate definitions. The primary experiments here rebuild the surfaces in physical coordinates and transform their points explicitly. A measured-versus-nominal excitation sensitivity calculation is provided, without conflating it with that metadata issue.

6.2 Independent comparison before merging

For an ideal constant-excitation slice e_g, the shifted original is

$$\widetilde{\psi}_{x,g}(s, q) = \psi_{x,g}(s - ke_g, q). \tag{31}$$

Compare two such functions at *identical* reduced coordinates inside their common supported region. If their values differ, one single-valued shared function cannot match both exactly. For two scalar values v_g, v_h, any common value has a maximum absolute discrepancy of at least $|v_g - v_h|/2$;

their unweighted mean minimizes the sum of squared discrepancies. This elementary observation concerns the compared surfaces, not an isolated physical error floor.

The measured excitation varies within each actual group, so we transform each group's measured points with its measured excitation and build a separate transformed interpolant. Original physical triangle connectivity is preserved. We compare each projected triangle's signed area with its original area and reject reversed facets. At a query, membership in all projected facets is counted; overlapping or otherwise ambiguous memberships are excluded for both compared surfaces. This avoids hiding a folded excitation graph by simply retriangulating its projected nodes. A 41-by-41 physical query grid on the lower-excitation original surface supplies anchor points and interpolated measured excitation. Those points are transformed for each method and queried on the neighboring higher-excitation surface at exactly the same reduced coordinates. The anchor's original flux is the comparison value. All seven reduced methods use the intersection of their supported physical anchors for that pair. Pairwise squared errors are pooled with their valid point counts; individual pair counts and both flux components are retained.

This procedure uses independently interpolated surfaces, before any shared raster is built. Its numerical path differs from the shared-table reconstruction and from local derivative fitting. It nevertheless retains measurement, excitation-graph and original-surface interpolation uncertainty. A nonzero difference cannot be assigned uniquely to physical model inadequacy, and a small difference between neighboring groups does not prove global compatibility of all groups.

Projected facets use straight edges through transformed vertices. For the nonlinear coordinates, this also approximates the curved image of an original facet. That discretization contribution is retained in the interpretation of the independent comparison; the two numerical procedures are complementary, not exact noise-free duplicates.

6.3 Parameter fitting without final-group leakage

Each outer fold removes one complete nominal excitation group. No point from that group is used to estimate a factor, fit a variable shift, set current scales or tune numerical coefficients. Within the remaining training groups, three interior groups spanning the available excitation range are selected by a fixed rule as inner folds. Each inner fold retains at most 120 deterministically spaced query records, inside the baseline model's support. Its predictor is fitted to the other inner groups. This makes an omitted group a prediction problem; it cannot receive a new free shift fitted to its own fluxes.

For C–G the criterion uses equal weights for the two flux components in Vs:

$$\mathcal{J} = \frac{1}{N_f} \sum_{f} \frac{1}{2n_f} \sum_{j \in f} \left(r_{d,j}^2 + r_{q,j}^2 \right) + \lambda \sum_{\ell \geq 2} (c_\ell / K)^2. \tag{32}$$

Here $K = \max(|k_A|, 1)$; $\lambda = 10^{-8}$ Vs^2 for D–G and zero for B/C. B uses d-flux alone. Inner groups receive equal weight, irrespective of sample count. A candidate losing a fixed inner query receives a 1 Vs residual for each missing component, preventing an optimizer from improving its score by discarding difficult points. This penalty is an optimization device, not an acceptance tolerance. Final errors include only explicitly supported points and report coverage.

B/C use bounded scalar minimization, refining the best two basins found in an initial bracket search over $[-0.5K, 2.5K]$. D–G start from C and use a bounded-coefficient simplex search, with coefficient magnitudes limited to $4K$, the monotonicity restriction for G, and a fixed evaluation budget. The objective can have local minima because the triangulation changes. Termination flags, evaluations, search bounds and every fold's parameters are saved. Failure to converge is reported as such; the best visited point is not claimed to be a unique global optimum. The hierarchy and regularization are specified before evaluating final groups.

For the all-data reconstruction, the same inner-fold selection rule fits coefficients before building

the table from all groups. Reconstruction uses training data and is not independent validation. Outer prediction refits the complete procedure for every removed group. Interior groups, boundary groups and unavailable extrapolation are distinguished. No extrapolation is enabled. A reduced model may predict a boundary group if its reduced coordinates remain inside training support, while H may have no support there. Such differences are reported rather than interpreted as matched-support extrapolation accuracy.

6.4 Metrics and aggregation

For a common set $\mathcal{V}$ of N supported physical points, define

$$\mathrm{RMSE}_x = \sqrt{\frac{1}{N}\sum_{j\in\mathcal{V}} r_{x,j}^2}, \tag{33}$$

$$J_{\mathrm{flux}} = \sqrt{\frac{1}{2N}\sum_{j\in\mathcal{V}} (r_{d,j}^2 + r_{q,j}^2)}, \tag{34}$$

$$M = \max_{j\in\mathcal{V},x\in\{d,q\}} |r_{x,j}|, \qquad Q_{95} = \mathrm{quantile}_{0.95}\left(\sqrt{r_d^2 + r_q^2}\right). \tag{35}$$

All four quantities have flux-linkage units. J_{flux} is the root mean square over two component samples per point; Q_{95} is a vector-error quantile and therefore has a different normalization. Coverage is the valid count divided by the eligible source count. Overall RMSE pools squared point errors and counts across groups; it is not the arithmetic mean of group RMSE values. Per-group errors, each model's own valid count, common-support counts, parameter counts and storage are exported. The main interior prediction comparison uses the common support of A–H; boundary comparisons use a separately identified reduced-model intersection and own-support results.

7 Results

7.1 The estimator and derivative evidence

Table 1: All-data constant factors in A/A. B uses fixed-q d-flux prediction; C uses joint flux prediction. The derivative LS column optimizes a different criterion.

Set	A	B	C	Derivative LS
1	13.4921	14.2412	14.3385	13.8374
2	13.5184	14.3506	14.1803	13.9514
3	14.7642	14.0315	13.9320	14.4482

The conventional factors are 13.4921, 13.5184 and 14.7642 for Sets 1–3. Their difference from the earlier saved numbers reflects a fresh common preprocessing run rather than reuse of historical results. The fixed-q estimate B and joint estimate C are identified independently. Differences among A, B, C and the derivative LS estimate cannot be attributed solely to q-current drift: the objectives, sampled points, surface approximations and weighting rules also differ.

Nonzero centroid departure confirms that mean contour coordinates are generally not points of the same constant-flux contour. The drift term in (7) is therefore not the only distinction between the implemented statistic and a fixed-q local derivative. The intersection mean is unweighted by contour length. Its sensitivity to mesh geometry and incomplete support is an algorithmic possibility; the present results do not isolate either mechanism as the physical cause of the measured discrepancies. The follow-up `k_factor_investigation` compares these effects without changing the conventional reference results.

Table 2: Contour diagnostics. Centroid RMS is the d-flux difference between a contour's coordinate centroid and its selected level; it is not a reconstruction RMSE. The final column quantifies centroid q-current drift.

Set	k_A	Centroid RMS in mVs	median $\lvert\Delta i_q/\Delta i_e\rvert$
1	13.4921	29.223	2.547
2	13.5184	28.854	3.712
3	14.7642	14.176	4.706

Table 3: Primary 80-neighbor derivative diagnostics. All residual norms are dimensionless ratios of inductance norms, not relative flux errors.

Set	k_{LS}	median η	ρ_A	ρ_{LS}	ρ_{local}
1	13.837	0.090	0.162	0.160	0.088
2	13.951	0.087	0.156	0.153	0.083
3	14.448	0.144	0.197	0.196	0.138

The global derivative factors are approximately 13.84, 13.95 and 14.45. Changing from A to the derivative-optimal constant only modestly reduces the aggregate derivative residual. Even a pointwise joint factor leaves a nonzero residual. The primary normalized-direction medians are approximately 0.090, 0.087 and 0.144. This is evidence of imperfect collinearity in the estimated sensitivities, with larger directional disagreement in Set 3. It is not a calibrated physical percentage error.

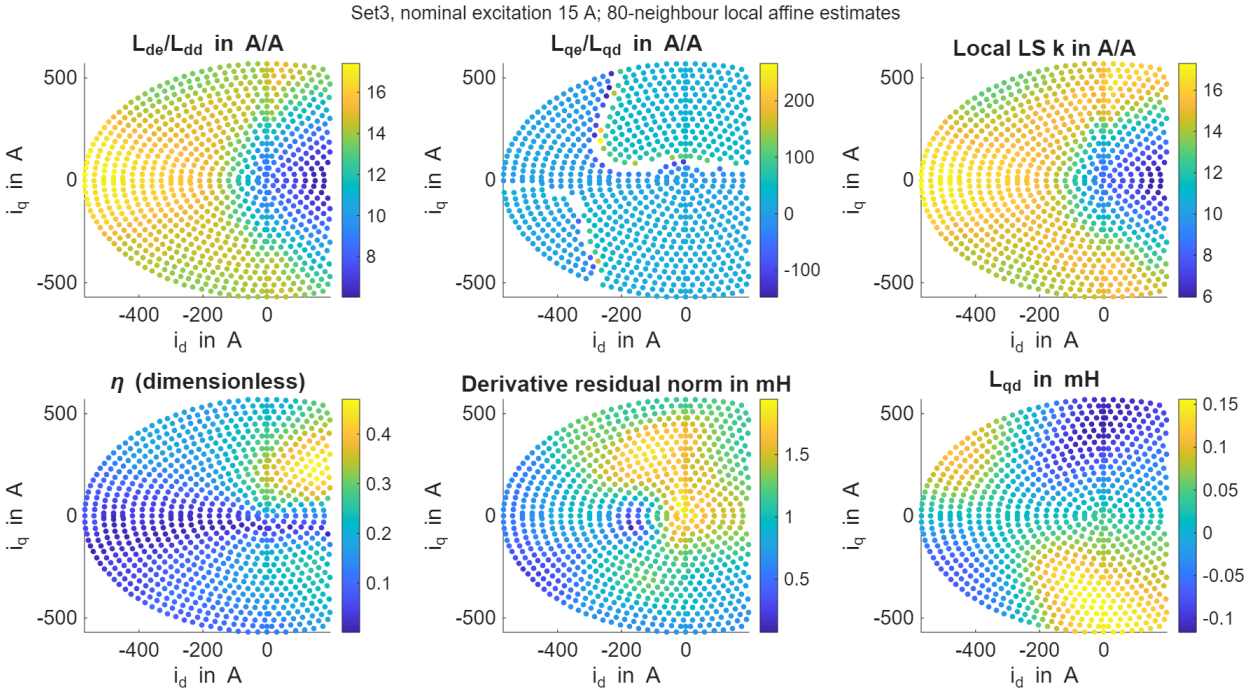


Figure 3: Representative Set 3 derivative maps. The separate q-flux ratio is poorly conditioned when its d-current sensitivity is small. Joint LS avoids dividing by that individual cross-inductance, while the determinant measure asks whether one local d-compensation direction can serve both outputs.

The neighborhood study retains the same source and coordinate convention and compares common derivative points. Its variation remains relevant when interpreting the numerical inductances; no confidence interval for exact physical derivatives is inferred.

Table 4: Derivative-neighborhood sensitivity on the same retained physical points within each dataset. The changes reflect numerical slope estimation, not a change in machine turns.

Set	Neighbours	Derivative LS	ρ
1	40	13.9539	0.1538
1	80	13.8831	0.1473
1	120	13.7939	0.1464
2	40	14.1177	0.1461
2	80	14.0148	0.1428
2	120	13.9344	0.1422
3	40	14.5383	0.1867
3	80	14.5119	0.1819
3	120	14.4667	0.1801

7.2 Independent transformed-surface compatibility

Table 5: Joint RMS disagreement of adjacent transformed original surfaces in mVs, pooled by valid pair counts on identical supported anchors across A–G. These are surface differences, not shared-table reconstruction errors.

Set	A	B	C	D	E	F	G
1	1.868	1.841	1.853	1.852	1.844	1.852	1.800
Common pair anchors: 13906/14677.							
2	1.820	1.801	1.784	1.774	1.782	1.787	1.751
Common pair anchors: 15076/15927.							
3	2.597	2.642	2.661	2.650	2.625	2.542	2.422
Common pair anchors: 10848/11712.							

The maps locate where separate groups ask a common table for different values. Improving the scalar factor can move and reduce some discrepancies, but a shared table cannot simultaneously reproduce distinct requested values at one coordinate. Pairwise incompatibility and derivative-direction disagreement are complementary evidence. They do not supply an isolated physical error floor because original-surface interpolation and measurement uncertainty remain present. The folding report records reversed projected facets and ambiguous query memberships; these queries are excluded rather than assigned an arbitrary surface value.

The pooled adjacent-surface disagreement for A is 1.868, 1.820 and 2.597 mVs in Sets 1–3; G changes these values to 1.800, 1.751 and 2.422 mVs. The larger Set 3 disagreement is qualitatively consistent with its larger derivative-direction diagnostic, while neither numerical procedure eliminates its own estimation uncertainty. These pairwise surface differences must not be subtracted from shared-table RMSEs: they use different queries and error definitions.

7.3 Shared-table reconstruction and missing-group prediction

The reconstruction rows include the entire storage-and-return operation. They use the source points again and therefore measure approximation of available data rather than prediction of new excitation groups. The prediction rows remove a full group, refit all parameters from the remaining data and then query the unseen group. These two rows of evidence must not be interchanged.

On the common interior prediction points, A has joint RMSE values of 5.998, 6.309 and 6.970 mVs in Sets 1–3. C gives 5.518, 5.736 and 6.996 mVs: optimizing a joint training criterion improves the first two sets but does not improve the unseen-group aggregate in Set 3. B also does not rank uniformly above or below C. These outcomes preclude attributing the remaining error solely to the conventional estimator or to moving q-current.

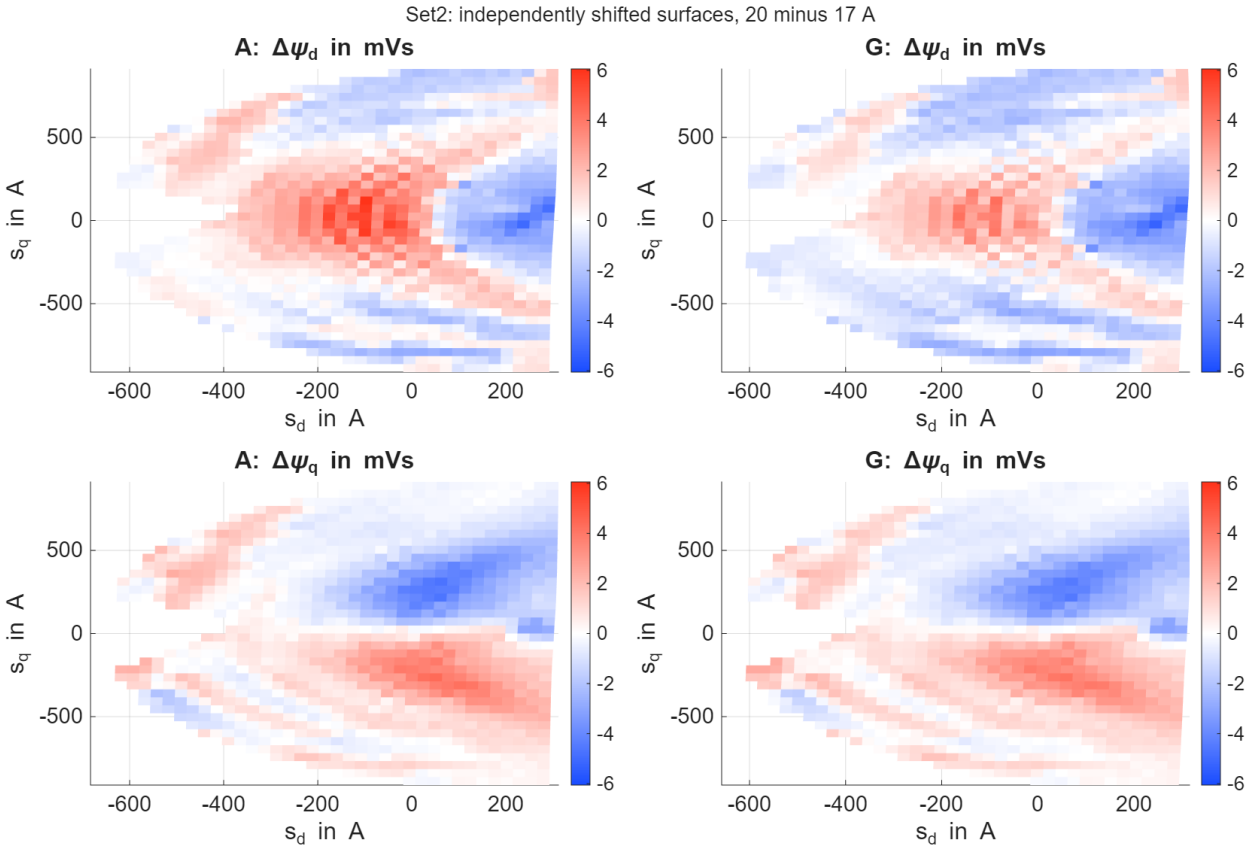


Figure 4: A representative adjacent excitation pair in Set 2. Each panel compares two independent original surfaces at the same reduced coordinates, before merging. Positive and negative differences are displayed with a common zero-centred scale. The methods use the same retained physical anchors.

Table 6: Set 1: errors on common support. R: reconstruction; P: interior whole-group prediction. Max: largest absolute component error; Q95: vector-error quantile.

Method	Mode	RMSE d	RMSE q	Joint	Max	Q95
A	R	5.239	4.041	4.678	45.87	15.66
B	R	4.643	3.955	4.313	32.29	13.48
C	R	4.624	3.873	4.265	29.43	13.21
D	R	4.626	3.876	4.267	29.46	13.28
E	R	4.520	3.864	4.205	29.29	12.98
F	R	4.587	3.862	4.240	29.38	13.16
G	R	4.474	3.956	4.223	29.70	13.19
H	R	0.682	0.301	0.527	4.57	1.57
A	P	6.557	5.380	5.998	50.34	19.10
B	P	5.927	5.223	5.586	31.29	16.87
C	P	5.914	5.092	5.518	31.41	16.48
D	P	6.001	5.109	5.573	31.22	16.70
E	P	5.785	5.099	5.453	32.75	16.60
F	P	5.896	5.052	5.490	30.53	16.47
G	P	5.800	5.210	5.513	33.95	16.97
H	P	2.362	0.898	1.787	12.09	5.68

Common points: reconstruction 5662/7476; interior prediction 5659/6348.

Table 7: Set 2: errors on common support. R: reconstruction; P: interior whole-group prediction. Max: largest absolute component error; Q95: vector-error quantile.

Method	Mode	RMSE d	RMSE q	Joint	Max	Q95
A	R	5.783	4.197	5.053	44.00	16.21
B	R	4.936	3.958	4.474	29.08	14.30
C	R	5.048	3.974	4.543	29.89	14.45
D	R	4.820	4.087	4.468	37.18	14.12
E	R	4.726	3.974	4.366	31.32	13.76
F	R	4.822	3.984	4.423	29.88	13.76
G	R	4.593	4.025	4.318	27.52	13.76
H	R	0.719	0.316	0.556	4.28	1.77
A	P	7.095	5.409	6.309	45.51	20.09
B	P	6.164	5.004	5.614	30.23	17.16
C	P	6.369	5.024	5.736	35.54	16.98
D	P	6.049	5.301	5.687	30.70	17.26
E	P	5.695	5.235	5.470	30.94	16.75
F	P	6.201	5.047	5.653	39.65	16.92
G	P	5.583	5.299	5.443	33.45	16.86
H	P	2.346	0.911	1.780	11.96	5.58

Common points: reconstruction 6175/8092; interior prediction 6175/6936.

Table 8: Set 3: errors on common support. R: reconstruction; P: interior whole-group prediction. Max: largest absolute component error; Q95: vector-error quantile.

Method	Mode	RMSE d	RMSE q	Joint	Max	Q95
A	R	5.462	5.469	5.466	34.75	16.34
B	R	4.953	5.602	5.287	30.59	15.29
C	R	4.880	5.605	5.255	30.05	15.24
D	R	4.926	5.618	5.283	30.17	15.38
E	R	4.897	5.401	5.155	31.88	14.95
F	R	4.820	5.173	5.000	26.68	14.87
G	R	4.009	5.628	4.886	26.06	14.60
H	R	0.409	0.210	0.325	2.02	0.85
A	P	6.899	7.041	6.970	35.94	19.93
B	P	6.629	7.199	6.920	33.81	18.63
C	P	6.837	7.151	6.996	32.47	19.01
D	P	6.328	7.080	6.714	32.82	18.40
E	P	6.616	7.055	6.839	32.36	18.58
F	P	6.147	6.890	6.529	33.84	18.17
G	P	5.580	7.171	6.425	30.91	17.84
H	P	1.898	0.883	1.480	8.33	3.91

Common points: reconstruction 6630/9022; interior prediction 6592/7264.

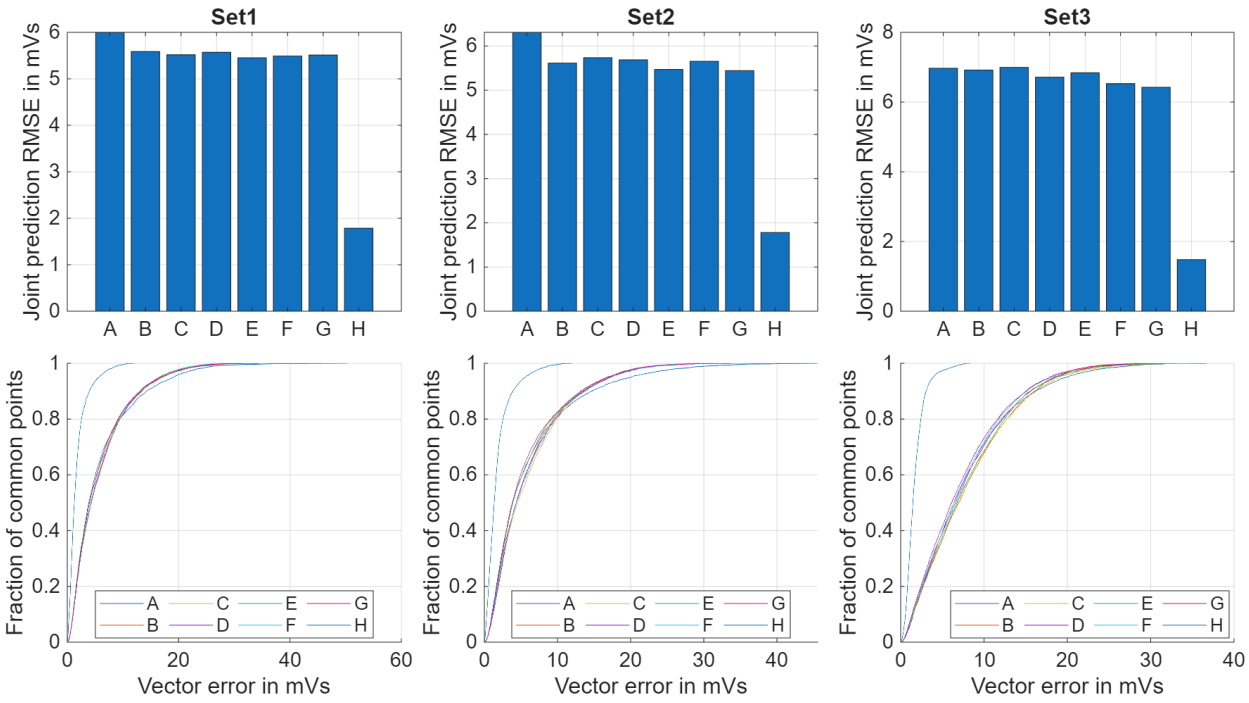


Figure 5: Overall method comparison. Upper panels: joint RMSE on the same interior whole-group prediction support for A–H. Lower panels: cumulative distributions of vector error on that support. The curves allow an error–coverage assessment without imposing an application-independent acceptance threshold.

E gives 5.453, 5.470 and 6.839 mVs; F gives 5.490, 5.653 and 6.529 mVs; G gives 5.513, 5.443 and 6.425 mVs. E is the best reduced candidate in Set 1, and G is best in Sets 2 and 3. G's advantage over E in Set 2 is only 0.027 mVs, while G is slightly worse than E in Set 1. In Set 3 its advantage over the simpler F is 0.104 mVs, with a lower d-flux error but a higher q-flux error. The ranking therefore depends on machine configuration and on the equal-component aggregate used here.

H gives 1.787, 1.780 and 1.480 mVs, substantially below every reduced candidate on the identical points. The common prediction coverages are 89.15%, 89.03% and 90.75% of the eligible interior points. This is a strong empirical accuracy advantage for retaining excitation in these data, although it includes the stated interpolation and storage effects and is not a proof against every possible two-coordinate representation.

The cumulative error curves are conditional on common supported points. To express a retained fraction relative to all eligible points, multiply the plotted fraction by the reported common coverage. Missing support is not a zero-error observation.

The coefficients describe restricted coordinate functions, not independent shifts fitted to each final group. Because each outer prediction uses its own training fit, the all-data coefficient table alone does not reproduce the prediction curves; the fold parameters are part of the saved numerical result. G's positive fixed-excitation Jacobian is checked at every held-out physical point as well as on its training box. The fold geometry report also counts points outside the training-current box; support in reduced coordinates alone is not evidence of interpolation in every physical current.

All B–F fits meet their stated numerical termination criterion. Four of the 40 G fits, counting the all-data and outer fits, reach a finite search limit instead. Their best visited coefficients are retained, but G's ranking is consequently a comparison of computed candidates, not a certified converged optimum. This further limits the case for preferring its additional complexity on the basis of very small RMSE differences.

Across all 37 held-out groups, the smallest evaluated G Jacobian is 0.759; none is nonpositive or

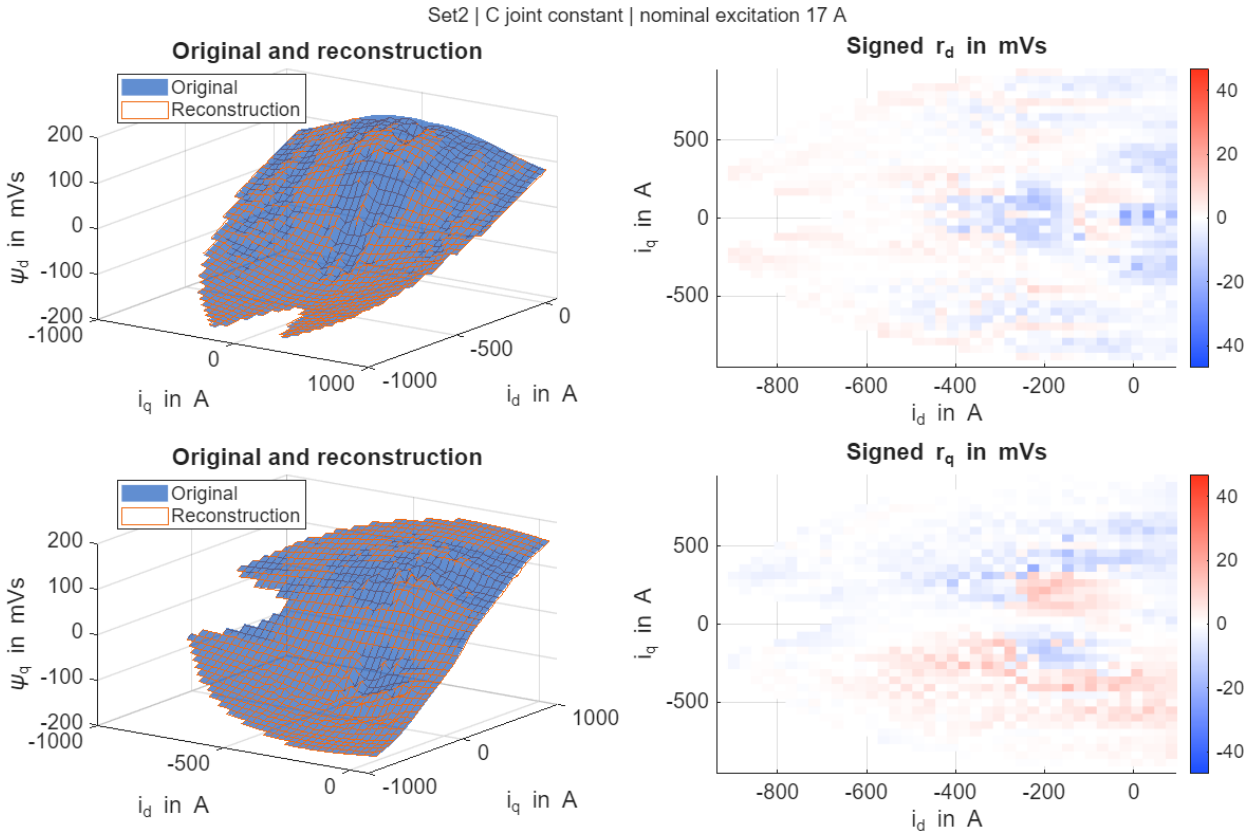


Figure 6: Representative Set 2 group for C. Each flux row compares the original and reconstruction in 3D at the left and shows the signed physical-current error at the right. Only supported parts of the original excitation graph are displayed. Numerical tables use point-based errors rather than this display grid.

Table 9: Fitted variable-shift coefficients in A/A. Empty coefficients are shown as NaN. The current scales come from the respective training dataset; coefficients of different datasets are not directly transferable.

Set	Method	c_1	c_2	c_3	c_4	c_5
1	D	14.3467	-0.0105	NaN	NaN	NaN
1	E	14.4339	-0.0381	-0.2185	NaN	NaN
1	F	14.3367	-0.0256	NaN	NaN	NaN
1	G	14.0170	0.1052	-0.6091	0.4575	0.5941
2	D	14.9016	-0.9459	NaN	NaN	NaN
2	E	14.7639	-0.3724	-0.1483	NaN	NaN
2	F	14.1870	-0.0918	NaN	NaN	NaN
2	G	14.1210	-0.2569	-0.4607	-0.0279	1.0622
3	D	14.0365	-0.1092	NaN	NaN	NaN
3	E	14.0883	0.0256	1.2640	NaN	NaN
3	F	14.6271	-0.7879	NaN	NaN	NaN
3	G	15.0134	-1.2991	0.0466	-0.2755	1.5915

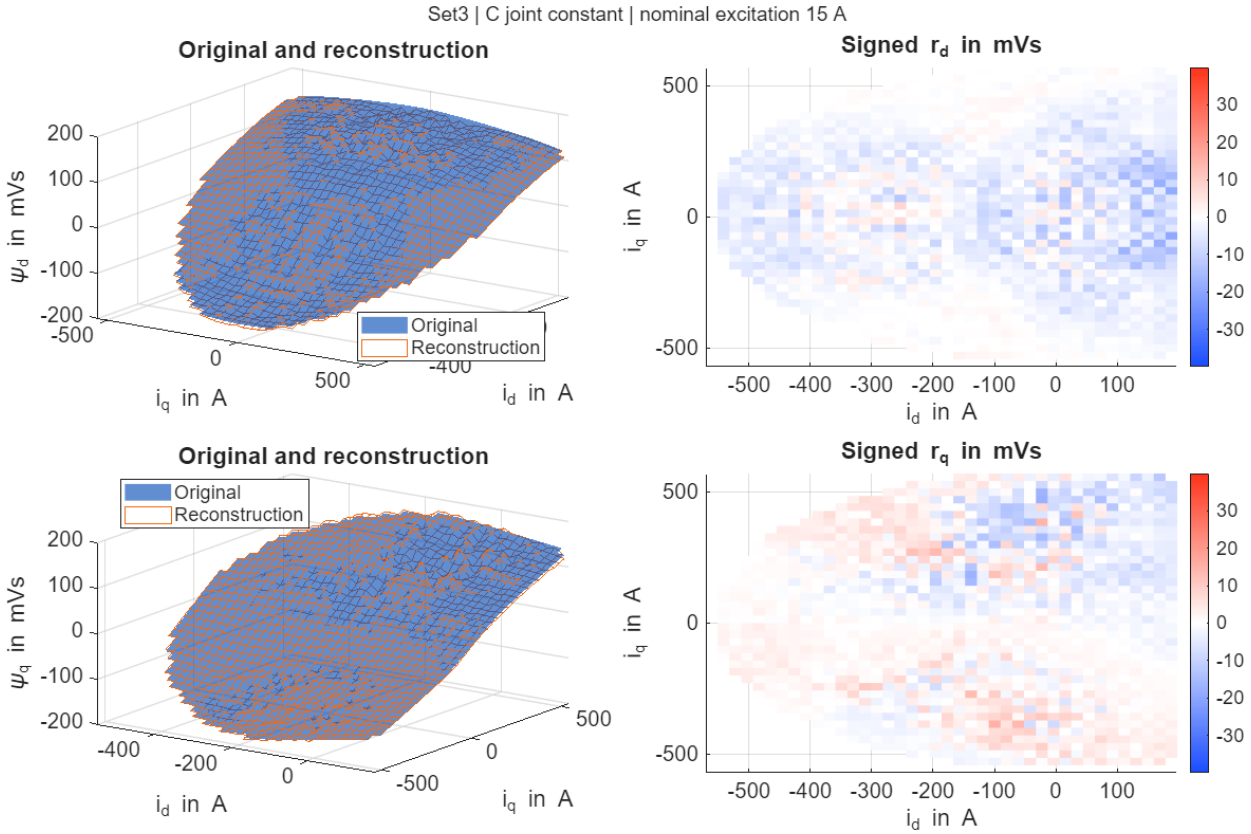


Figure 7: The corresponding representative view for the separate machine in Set 3. The atlas supplies the same comparison for every excitation group and method.

Table 10: Number of fitting runs with a nonpositive termination flag, including the all-data fit and every outer fit. Finite-budget outcomes are retained transparently rather than called converged optima.

Set	B	C	D	E	F	G	Runs/model
1	0	0	0	0	0	2	14
2	0	0	0	0	0	1	15
3	0	0	0	0	0	1	11

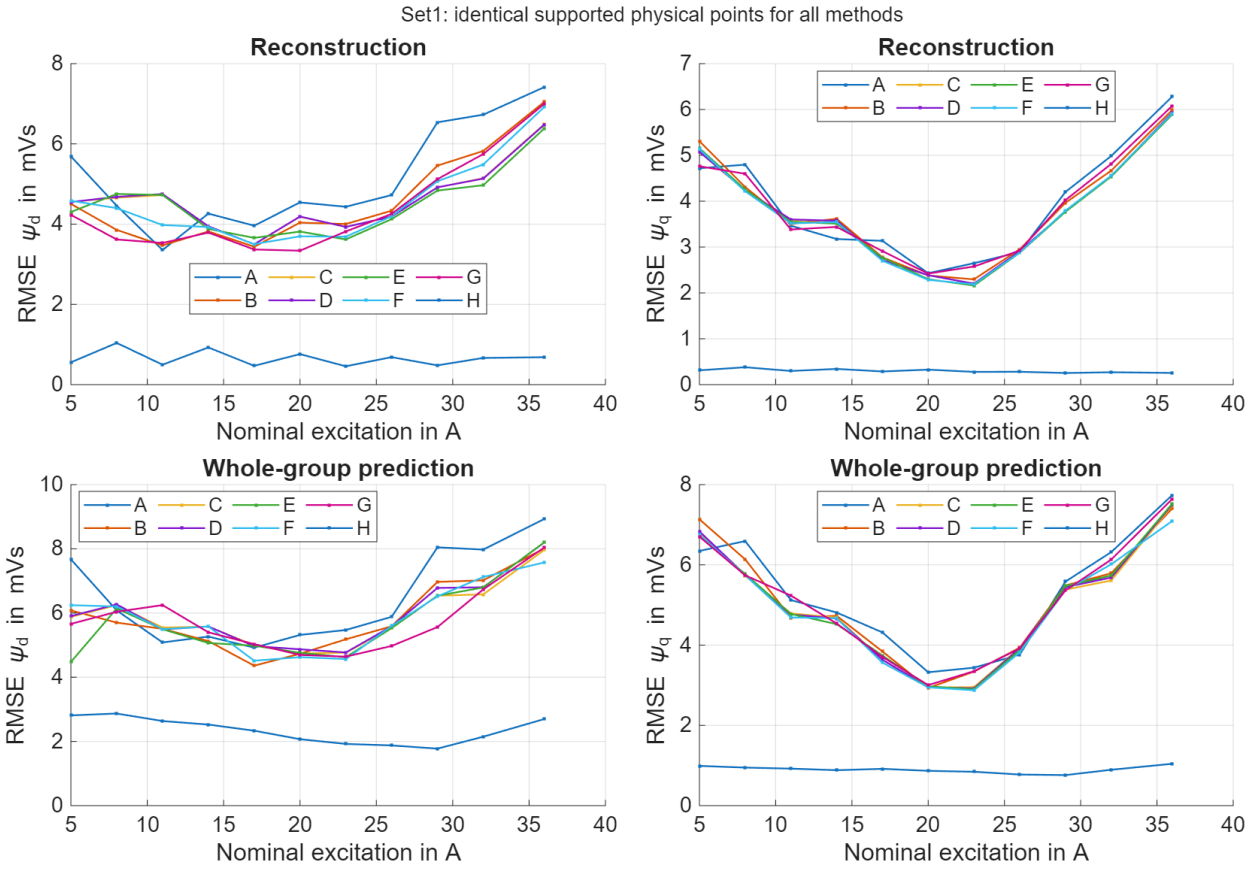


Figure 8: Set 1 per-group RMSE. Upper panels: reconstruction. Lower panels: interior whole-group prediction. A missing boundary marker denotes absence from the all-model interior comparison, not an observed zero error. Boundary results are reported separately.

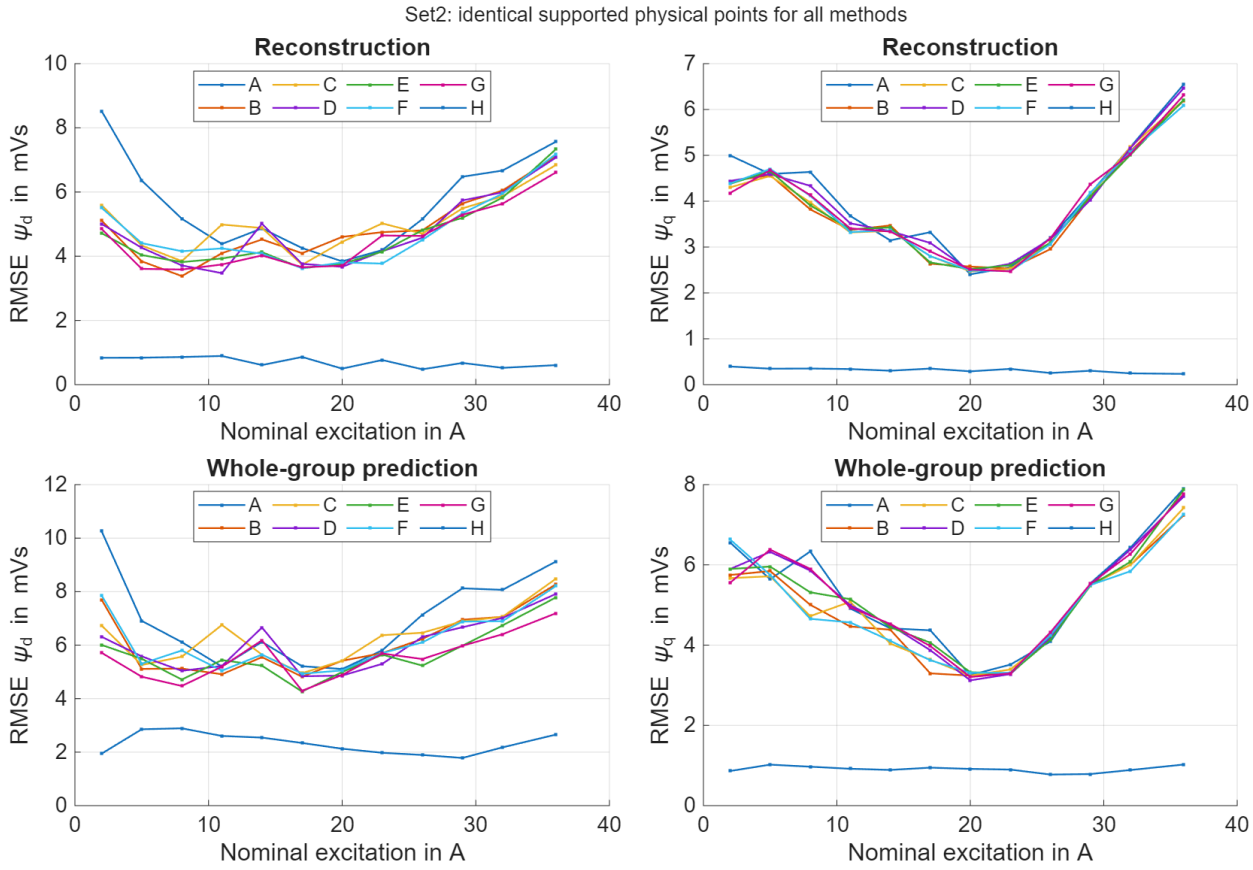


Figure 9: Set 2 per-group RMSE with the same definitions and common-support policy.

below the training margin of 0.25. This confirms the fixed-other-currents monotonicity check at the evaluated points, while the separate empirical excitation-graph folding check remains necessary.

7.4 Boundary groups and unsupported extrapolation

Boundary removal is more demanding than interpolation between retained excitation groups. H can lose all support when no remaining point surrounds the query in physical three-current space. Reduced coordinates may remain supported even though excitation lies outside the observed training range. The reported boundary errors assess only those available measured points; they do not validate unrestricted excitation extrapolation or an unmeasured operating range.

7.5 Numerical controls and excitation-reference sensitivity

Raw and normalized axes can generate different triangulations even with identical source values. Differences between a normalized 3D reference and an unnormalized 2D reduction would mix coordinate-model and numerical effects. Here both are normalized in the primary comparison, and the four-way control makes the scaling dependence visible. Changing raster density quantifies the storage contribution for A without identifying a unique physical cause of the remaining error. RMSE differences between these rows are not subtracted to estimate a structural error component.

Within-group excitation scatter therefore has a concrete coordinate consequence. This calculation keeps the same shared table and changes only the query. It is separate from the fixed-reference convention of a relabelled historical single-group table. The source audit confirms the actual convention of each available historical export rather than assuming that every export already carries current metadata.

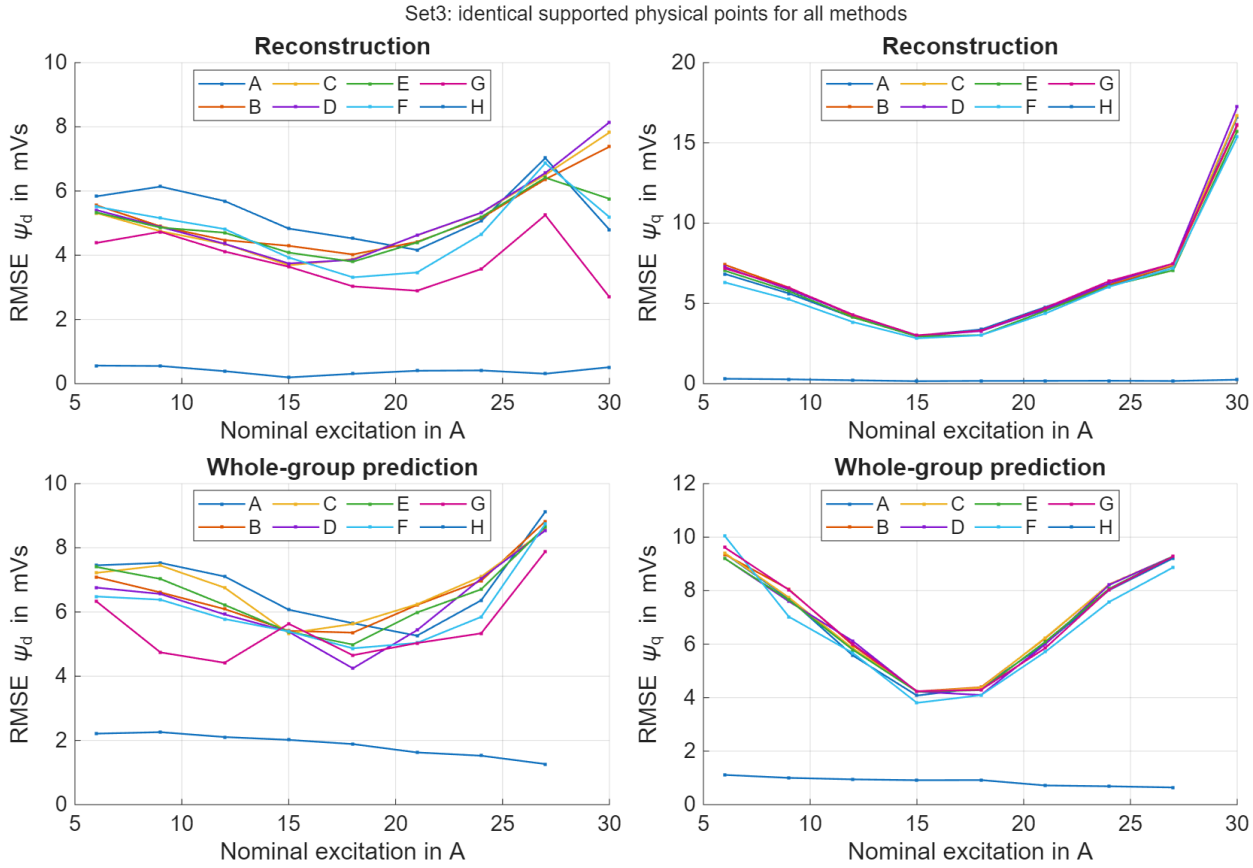


Figure 10: Set 3 per-group RMSE. Each curve's parameters were identified independently for this machine configuration.

Table 11: Boundary-group prediction for representative methods. C/G use the reduced-model common support. H uses its own support and is therefore not a matched-support accuracy comparison with C/G. Excitation is nominal, in A. Full A–H boundary metrics are supplied per group in the numerical material.

Set	Excitation	Method	Joint in mVs	Points	Coverage
1	2	C	7.142	509	0.881
1	2	G	7.187	509	0.881
1	2	H	NaN	0	0.000
1	40	C	11.127	455	0.827
1	40	G	11.986	455	0.827
1	40	H	NaN	0	0.000
2	0	C	6.554	515	0.891
2	0	G	7.475	515	0.891
2	0	H	NaN	0	0.000
2	40	C	13.336	465	0.804
2	40	G	10.981	465	0.804
2	40	H	NaN	0	0.000
3	3	C	10.238	811	0.874
3	3	G	9.568	811	0.874
3	3	H	NaN	0	0.000
3	30	C	11.805	723	0.871
3	30	G	11.115	723	0.871
3	30	H	NaN	0	0.000

Table 12: Scaling and raster controls. The four scaling rows use one common physical mask per dataset. The three raster rows use a separate common mask across the three raster resolutions. Counts are therefore not interchangeable between controls. Errors are in mVs.

Set	Control	RMSE d	RMSE q	Points
1	A normalized	5.244	4.039	5625
1	A raw axes	5.341	4.089	5625
1	H normalized	0.683	0.301	5625
1	H raw axes	0.771	0.363	5625
1	A raster 41	6.716	5.527	7129
1	A raster 81	5.628	4.406	7129
1	A raster 161	4.356	3.236	7129
2	A normalized	5.796	4.195	6131
2	A raw axes	5.836	4.192	6131
2	H normalized	0.721	0.316	6131
2	H raw axes	0.861	0.394	6131
2	A raster 41	7.114	5.501	7729
2	A raster 81	6.279	4.498	7729
2	A raster 161	4.622	3.241	7729
3	A normalized	5.363	5.456	6607
3	A raw axes	5.370	5.459	6607
3	H normalized	0.409	0.210	6607
3	H raw axes	0.762	0.418	6607
3	A raster 41	6.533	6.657	8667
3	A raster 81	5.837	5.934	8667
3	A raster 161	4.384	4.264	8667

Table 13: Sensitivity of the A table output to replacing measured excitation by the nominal group label at the query. These are differences between two returned predictions, not measurement errors.

Set	RMS d in mVs	RMS q in mVs	Points
1	2.042	1.014	5662
2	2.024	0.977	6175
3	2.556	1.577	6630

Three small analytic controls were evaluated directly. Moving between distinct physical points on a fibre gives zero discrepancy for an exact constant shift; the integrated variable shift agrees to about 1.1×10^{-16} Vs. Replacing the integrated shift by $i_e k_{\mathrm{diff}}$ in that example gives a maximum error of 0.27 Vs. A deliberately incompatible example gives the same (s, q) but q-flux values separated by 2 Vs. These controls illustrate the mathematical distinctions; their arbitrary scales are not machine-accuracy benchmarks.

8 Physical interpretation and remaining uncertainty

8.1 Coordinate compression is not complete machine equivalence

For the constant d-shift, even an exact stator-flux reduction leaves the physical currents $i_d = s - k i_e$ and $i_q = q$. With the conventional three-phase amplitude-invariant dq torque identity,

$$T_e = \tfrac{3}{2} p (\psi_d i_q - \psi_q i_d) \tag{36}$$

$$= \tfrac{3}{2} p \left[F_d(s, q) q - F_q(s, q) s + k i_e F_q(s, q) \right]. \tag{37}$$

Substituting s directly for physical i_d would remove the final term incorrectly. Likewise, the resistive voltage is $R_s(s - k i_e)$, not $R_s s$. The excitation circuit remains a separate physical circuit. For variable coordinates, the complete derivatives in (27) must be retained in dynamic calculations. A compact stator-flux table does not establish identical torque, voltage limits or controller dynamics to a permanent-magnet machine.

For F, the inverse is $i_d = s_d - k_d i_e$ and $i_q = s_q - k_q i_e$. Both recovered physical currents, rather than the two table axes themselves, must enter the torque and resistive-voltage expressions.

8.2 What the experiments separate, and what they do not

Different k estimators expose sensitivity to the fitting criterion and contour construction. Persistent derivative-direction residuals and independent-surface disagreement expose departures from the reduction constraint in the estimated data. Changing coordinate normalization or raster density exposes numerical sensitivity. Raw-to-export comparisons expose changes in preprocessing. These are distinct diagnostics, not additive causal components.

In particular, subtracting two RMSE values does not estimate a physical model-error contribution. The errors can be correlated, the interpolation geometry can change, and coverage can change. Measurement calibration, period averaging, repeat averaging, nonstationarity, thermal drift and fitted-surface error have not been separately quantified. Therefore neither the derivative residuals nor the surface differences constitute an uncertainty-calibrated rejection of an underlying physical relation. The full three-current reference is also subject to these limitations.

Whole-group prediction removes a significant source of fitting leakage but does not create a new independent measurement campaign. The finite number of groups, fixed polynomial hierarchy and finite optimizer budget limit the strength of any ranking. The comparison does not claim that all possible nonlinear two-coordinate representations have been ruled out. Nor does a small all-data reconstruction error demonstrate lossless reduction.

9 Recommendation and conclusion

If a two-coordinate table is required and both stator axes may be shifted, F is the most convincing simple k-based compromise in this comparison. It retains two constant parameters and a globally simple linear coordinate map, improves joint interior prediction over A in every dataset, and is close to the best restricted nonlinear candidates. Its joint RMSE reductions relative to A are

approximately 8.5%, 10.4% and 6.3%. These percentages compare the defined flux-error statistic; they are not the derivative-angle measure. No application acceptance threshold is inferred.

E deserves consideration when the q-axis must remain the physical q-current and performance on the first machine configuration is the priority. It has the lowest reduced-model error in Set 1 and nearly matches the five-parameter G in Set 2. That result does not transfer unchanged to Set 3, where the simple two-axis F performs better. G has the lowest reduced-model joint error in Sets 2 and 3, but its extra coefficients, nonlinear inverse conditions and component trade-offs buy only a small improvement over the simpler nearest competitors. A remains a useful named baseline, while C is a better-founded constant d-shift fitting criterion rather than a universally superior predictor.

When flux accuracy dominates storage, H is preferable within the supported measured domain. It reduces the joint missing-group RMSE to approximately 1.5–1.8 mVs compared with approximately 5.4–7.0 mVs for the reduced candidates. This advantage costs an explicit excitation axis: the realized full tables have 137,781 nodes versus 6,561 for the reduced tables, or about 2.20 MB versus 0.105 MB for two double-precision flux arrays before masks and axes. It is an empirical reference with finite coverage, not an exact physical truth or an extrapolation guarantee.

The mathematical conclusion is unchanged by estimator improvements: a common constant d-shift requires both equations (15), with the stated global fibre conditions. A variable pure d-shift cannot compensate an absent common local direction merely by adding scalar parameters. The independent-surface and derivative checks address that compatibility from different numerical paths, while reconstruction and whole-group prediction quantify its practical consequences. Temperature, sensor and preprocessing uncertainty prevent a unique physical error partition. The recommended reduced model is therefore an approximation to be assessed against an application's own flux-error requirements, not a lossless replacement established by these experiments.

A Reproducibility and companion material

The research scripts rebuild the point exports, fit A–H, retain all outer-fold parameters, and generate the numerical tables and figures from their MAT results. The companion execution note records the source provenance, software version, exact commands, optimization settings, historical-coordinate checks and review-to-change mapping. Reproduction requires the acquisition MAT files; productive GUI changes are not required. Large historical exports are used only for a separately identified provenance comparison.

The atlas contains one page for every dataset, nominal excitation group and model, showing both flux components. Each row places a blue original surface and an orange reconstruction on the left and a signed physical-current error map on the right. A 41-by-41 display grid is supported only where the original surface and the reconstructed table are available. Measured excitation is interpolated on the original group graph. The error scale is zero-centred and fixed within each dataset across methods and groups. This display interpolation is separate from the point-based numerical metrics.

The d-flux 3D view places q-current on its first displayed axis and d-current on its second to show the surface more clearly. Error maps retain the physical d/q order. Axis units are written as "in A" and "in mVs". Atlas pages use one shared LaTeX template and nested loops over excitation groups and methods; the generated page list contains only the three dataset definitions.